

AI-Based Road Surface Defect Monitoring System

Smart City Road Management

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Presentation Outline

- 1 Problem Statement
- 2 Research Gaps & Challenges
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- 4 System Architecture
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- 6 Summary

Problem Statement

Urban Road Challenges

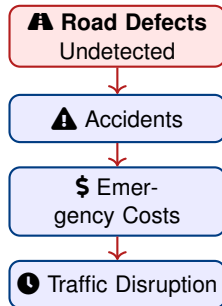
- Municipalities manage **large infrastructure** networks.
- Defects: **potholes, cracks, edge erosion, drainage**.
- Affects vehicle safety, travel quality, & maintenance budget.

Current Inspection Failures

- Manual periodic surveys $\Rightarrow$ **incomplete visibility**.
- Defects go undetected between cycles.
- Leads to accidents, emergency repairs, disruptions.
- No spatially-indexed, severity-based defect database.

Requirement

An **automated, continuous, data-driven** road defect detection and prioritization system for smart city maintenance planning.



Research Gaps & Key Challenges



Imaging Challenges

- **Light & Shadow Variation**
Inconsistent illumination degrades features.
- **Weather Changes**
Rain/fog alter surface appearance.
- **Blurry Images**
Vehicle vibration causes motion blur.



System-Level Gaps

- **Precision vs. Latency**
Models too slow for real-time edge.
- **No Priority System**
No severity-indexed defect triage.
- **No Geo-tagged Database**
Absence of spatially-indexed records.

Gap Summary

Challenge	Prior Work	This Work
Real-time edge inference	×	✓
Weather robustness	Partial	✓
Defect prioritisation	×	✓
Geo-tagged database	×	✓

Proposed Solutions

Dynamic Frame Dropping

- Frame rate adapts to **vehicle speed**.
- Reduces redundant data ingestion.
- Preserves frames at high speeds.
- Reduces overall compute load.

Confidence-Based Routing

- High-confidence: **metadata saved only**.
- Low-confidence: sent to cloud for re-inference.
- Reduces bandwidth by **~60–70%**.

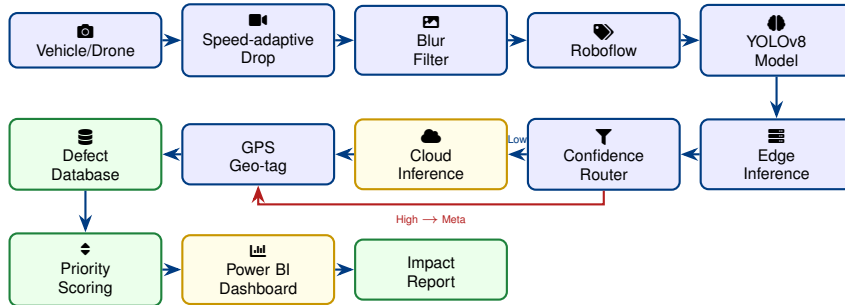
Blur Mitigation Pipeline

- Laplacian variance filters **blurry frames**.
- Temporal frame averaging for static zones.
- Deblur preprocessing for critical images.

Defect Priority Scoring

- Tiers: **Critical / Moderate**.
- Score = defect class + area + frequency.
- Dashboards with **geo-ranked queues**.

System Architecture Overview



Architecture: Milestone Breakdown

M1 – Data Collection Architecture

- Vehicle & drone camera survey protocols.
- Roboflow annotation & severity tiers.
- Overall inference integration plan.

M4 – Power BI Dashboard

- Road health scores by zone/corridor.
- Defect severity heatmaps.
- Maintenance priority lists.

M2 – Annotation & Model Training

- Annotate images: defect types & severities.
- Train YOLOv8 object detection model.
- Evaluate on representative test samples.

M5 – Urban Planning Impact Report

- Demonstrate improved prioritization.
- Estimate reduction in emergency costs vs. planned maintenance.

M3 – Batch Inference & Geo-Tagging

- Detect defects from survey images.
- Associate GPS metadata to detections.
- Output: geo-tagged defect database.

Key Deliverable Stack

Roboflow → YOLOv8 → Python Pipeline
→ Geo-tagged DB → Power BI

Dataset Details & Standards — Overview

Dataset 1: Vehicle-Mounted Camera (RDD-2022)

- > 47,000 ground-level road images from moving vehicles.
- Collected across Japan, India, Norway, USA, China.
- Captures close-up texture: cracks, potholes, surface wear.
- Trained independently on ground-perspective features.

[\[Drone Survey\]](#)

Dataset 2: Drone Survey (Top-Down)

- High-resolution aerial imagery from UAV surveys.
- Wider field-of-view: captures road edges, drainage, manhole covers, and large-scale cracking patterns.
- Trained separately on top-down perspective

Defect Categories

Code	Type	Source
D00	Severe Cracks	Both
D10	Mild Cracks	Both
D20	Severe potholes	Both
D40	Mild Potholes	Both
D50	Manholes	Drone only
D60	Smooth roads	Both

Database Output Schema

Field	Type
Timestamp	ISO 8601
GPS Coords	Lat, Lon
Defect Class	Enum (D00–D60)
Severity Tier	Critical/Mod/Minor
Confidence	Float [0, 1]

Dataset Samples — Vehicle-Mounted Camera

Ground-Level Samples (RCSD) — Captured from Moving Vehicle



Smooth Road



Crack — Mild



Crack — Severe



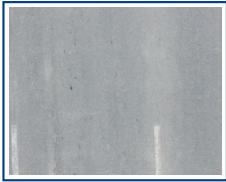
Pothole — Mild



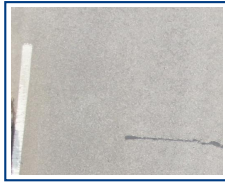
Pothole — Severe

Dataset Samples — Drone Survey

Drone Samples (RDD-2022) — Captured from Drones



Smooth Road



Crack — Mild



Crack — Severe



Pothole — Mild



Pothole — Severe



Manhole

Summary & Conclusion

What We Address

- ✓ Real-time edge inference
- ✓ Weather & illumination robustness
- ✓ Blur rejection before model inference
- ✓ Confidence-based bandwidth routing
- ✓ Severity-ranked priority queue
- ✓ Geo-tagged DB + Power BI analytics

Expected Outcomes

- Reduced emergency repair costs.
- Higher coverage than manual cycles.
- Data-driven budget allocation.

**Smart City Road
Management**

Real-time Detection

Priority Scoring

Power BI Dashboard

Thank You

Questions & Discussion

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